

Concept:

In IEEE- 754 single precision format, a floating-point number is represented in 32 bits.

Sign bit (MSB)	Biased Exponent (E') (8 bits)	Normalized Mantissa (M') (23 bits)
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Sign bit value 0 means positive number, and 1 means a negative number.

The floating-point number can be obtained by formula: $\pm 1. M \times 2^{(E-127)}$

Data:

Content of R1: 0x 42200000 (0x means Hexadecimal notation)

Content of R2: 0x C1200000

Calculation:

Content of R1 in Hex (0x) is 42200000. After converting into binary, it can be represented in IEEE- 754 format as:

0	100 0010 0	010 0000 0000 0000 0000 0000
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Sign bit is 0 i.e. the number is positive

Biased Exponent (E') = 100 0010 0 = 132

Normalized Mantissa (M) = 010 0000 0000 0000 0000 0000 = .25

Therefore, the number in register R1 = $+ 1.25 \times 2^{(132-127)} = 1.25 \times 32 = 40$

Content of R2 in Hex (0x) is C1200000. After converting into binary, it can be represented in IEEE- 754 format as:

1	100 0001 0	010 0000 0000 0000 0000 0000
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Sign bit is 1 i.e. the number is negative

Biased Exponent (E') = 100 0001 0 = 130

Normalized Mantissa (M) = 010 0000 0000 0000 0000 0000 = .25

Therefore, the number in register R1 = $-1.25 * 2^{(130-127)} = -1.25 * 8 = -10$

$R3 = R1/R2 = 40/-10 = -4$

Since the number is negative, Sign bit (MSB) = 1

Converting 4 into binary of a floating point gives: $(100.0)_2$

Representing it into normalized form gives: $(1.000000....) \times 2^2$

Therefore, Mantissa is 23 bits of all 0s

Biased Exponent (E') = $E + 127 = 2 + 127 = 129 = (10000001)_2$

It can be represented in IEEE- 754 format as:

1	100 0000 1	000 0000 0000 0000 0000 0000
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Converting it into Hex format gives: 0x C0800000